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Predictive Resource Optimization Platform (PROP) - KPMG Case Challenge Submission



🏆 Team 7

Team Member	GitHub	Linkedin	Background
Minh	@vibqetowi	hminh-software-eng	Software Engineering, previous experience in software development and project management
Carter	@carterj-c	cartercameronfina	Previously Mechanical Engineering now in Finance, previous experience in aerospace engineering
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🎯 Project Overview

Team 7 presents its submission for the KPMG case challenge: the **Predictive Resource Optimization Platform (PROP)**, focused on optimizing consulting assignment allocation. Leveraging our team's expertise in engineering, economics, project management, and finance, we have developed a solution addressing critical resource allocation challenges common within professional services organizations. While our primary backgrounds are not in direct consulting delivery, our technical skills and project coordination experience inform this solution. PROP is presented as a production-ready Proof of Concept (PoC) dashboard designed to implement enhanced resource management principles, adapting concepts from Earned Value Management (EVM) for the consulting context, to optimize resource allocation across engagements.

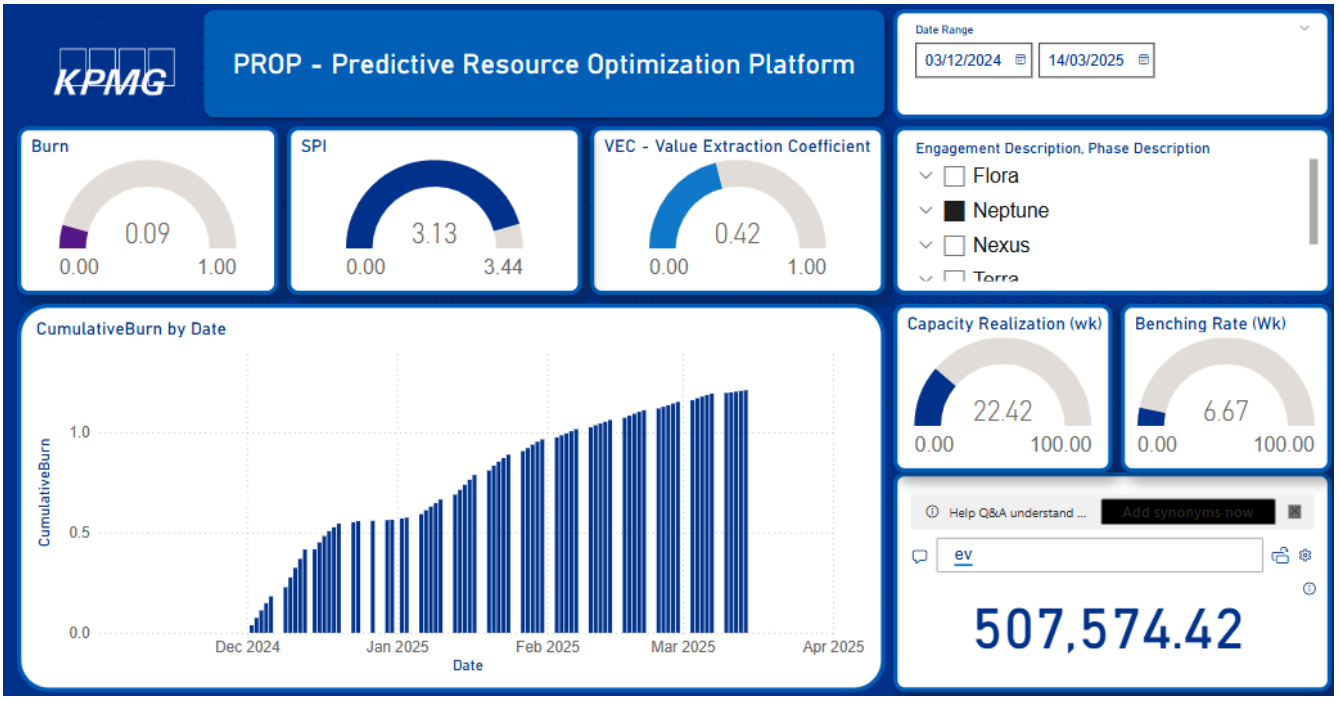
This dashboard is specifically engineered to address critical business needs prevalent in consulting practices:

- Optimizing consultant utilization and minimizing bench time.

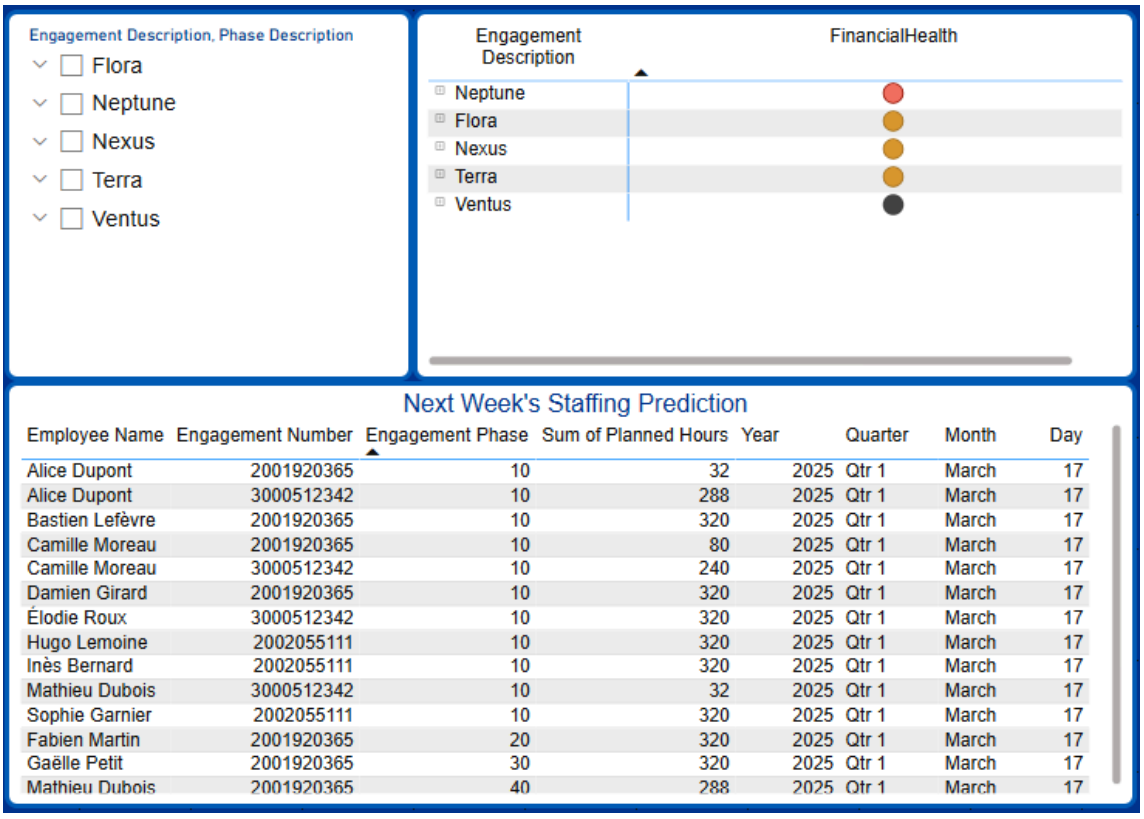
- Maintaining engagement schedules effectively.
- Maximizing profitability through efficient resource allocation.
- **Reducing management overhead by approximately 50%**: Significantly decreases the time managers spend on **manual data compilation and internal capacity planning reports**, allowing them to focus on billable client work – directly improving ROI through both cost reduction and revenue enhancement.

Platform Showcase

Main Dashboard



Next Week's Staffing Allocation Prediction



Employee Name	Engagement Number	Engagement Phase	Sum of Planned Hours	Year	Quarter	Month	Day
Alice Dupont	2001920365	10	32	2025	Qtr 1	March	17
Alice Dupont	3000512342	10	288	2025	Qtr 1	March	17
Bastien Lefèvre	2001920365	10	320	2025	Qtr 1	March	17
Camille Moreau	2001920365	10	80	2025	Qtr 1	March	17
Camille Moreau	3000512342	10	240	2025	Qtr 1	March	17
Damien Girard	2001920365	10	320	2025	Qtr 1	March	17
Élodie Roux	3000512342	10	320	2025	Qtr 1	March	17
Hugo Lemoine	2002055111	10	320	2025	Qtr 1	March	17
Inès Bernard	2002055111	10	320	2025	Qtr 1	March	17
Mathieu Dubois	3000512342	10	32	2025	Qtr 1	March	17
Sophie Garnier	2002055111	10	320	2025	Qtr 1	March	17
Fabien Martin	2001920365	20	320	2025	Qtr 1	March	17
Gaëlle Petit	2001920365	30	320	2025	Qtr 1	March	17
Mathieu Dubois	2001920365	40	288	2025	Qtr 1	March	17

🌟 Key Capabilities

PROP delivers significant advantages across several key operational areas:

1. Financial Performance & Profitability

- **Enhanced Profitability Visibility:** Provides clear, real-time insight into engagement profitability.
- **Rapid Financial Analysis:** Delivers instant financial interpretations, including the Value Extraction Coefficient (VEC), Budget Burn Rate, and Schedule Performance Index (SPI).

2. Resource Management & Optimization

- **Optimized Resource Planning:** Supports improved work-life balance through realistic and effective capacity planning.
- **Intelligent Consultant Allocation:** Features an algorithm for recommending optimized consultant assignments based on maximizing value and aligning with effective staffing patterns.
- **Comprehensive Resource Tracking:** Monitors key resource utilization metrics, including bench time and overall capacity utilization.
- **Integrated Absence Management:** Incorporates vacation and time-off tracking to prevent resource allocation conflicts and ensure realistic capacity assessments.

3. Project Execution & Performance Management

- **Adapted EVM Calculations:** Automatically calculates crucial metrics adapted from Earned Value Management (EVM). **Recognizing the inherent challenge of reliably measuring percentage completion (% Complete) for knowledge work typical in consulting engagements – where progress is often non-linear, milestones can be fluid, and objective quantification is difficult – this PoC adopts a pragmatic approach.** Instead of relying on subjective % Complete estimates, it calculates Earned Value (EV) based on Actual Cost (AC) adjusted by an efficiency factor (see Appendix 4). This provides a more grounded method for tracking progress against plan within the consulting context, alongside Planned Value (PV) and Schedule Performance Index (SPI).
- **Actionable Performance Insights:** Offers clear visualizations of engagement performance for informed decision-making.
- **Proactive Variance Management:** Enables effective monitoring of schedule and cost variances to maintain engagement control.
- **Improved Client Satisfaction:** Facilitates consistent adherence to project deadlines, enhancing client satisfaction (an outcome directly linked to effective project execution).

💼 Business Objectives and Optimization Targets

Our solution focuses on key business metrics that drive profitability and efficiency in consulting organizations: maximizing both budget utilization and maintaining target chargeout rates (captured by VEC), while ensuring timely delivery (SPI) and optimal resource usage (Benching Rate, Capacity Utilization Rate), **while addressing critical operational pain points.** These include the **significant administrative burden on resource managers** who often spend considerable time manually aggregating data for planning and reporting, as well as the **inherent difficulty in applying traditional project progress metrics, like percentage completion, to the often non-linear and knowledge-based nature of consulting work.** PROP aims to automate data aggregation and provide more suitable metrics like VEC and adapted SPI to overcome these challenges.

Value Extraction Coefficient (VEC)

Traditional project management metrics often fail to capture the dual financial objectives of consulting: utilizing the budget entirely (not leaving money on the table) while maximizing the value realized from chargeout rates and resource mix. Our Value Extraction Coefficient (VEC) provides executives with a consolidated measure of *financial realization efficiency* relative to the budget for an engagement or phase.

The VEC is calculated as follows (shown for a specific phase ph):

$$VEC_{ph} = \frac{1}{BAC_{ph}} \times \sum_{t=1}^{n} [AC_t \times d_t \times ea_c \times eo_c]$$

Where:

- BAC_{ph}: Budget At Completion for phase ph. The $\frac{1}{BAC_{ph}}$ term normalizes the extracted value against the phase budget baseline. (VEC can also be calculated at the engagement level).
- $\sum_{t=1}^{n}$: Summation over all individual financial transactions t (typically timesheet entries) contributing to the phase's actual cost up to the point of measurement.
- AC_t: Actual Billed Amount (or Cost equivalent) for transaction t.
- d_t: Rate Efficiency Factor for transaction t, measuring how the actual chargeout rate used compares to the standard rate for the consultant c associated with transaction t.

$$d_t = \frac{\text{Chargeout}_t}{\text{StandardChargeout}_c}$$

- ea_c: External Adjustment Factor for consultant c (e.g., 1.0 for internal, 0.9 for external).

$$ea_c = 1 - 0.1 \times \text{isExternal}_c$$

- eo_c: Onboarding/New Hire Adjustment Factor for consultant c (e.g., 1.0 for experienced, 0.8 for new).

$$eo_c = 1 - 0.2 \times \text{isNew}_c$$

Interpreting Financial Health (using VEC and Burn)

Budget Burn for a phase is calculated as $\text{Burn}_{ph} = \frac{\sum AC_t}{BAC_{ph}}$. VEC and Burn provide a nuanced view of financial health:

	VEC > Burn (Premium Realization)	**VEC ≈ Burn (Standard Realization)**	**VEC < Burn (Value Dilution)**
Burn < 1	Excellent	On Track	Warning
Burn ≈ 1	Excellent	On Track	Warning
Burn > 1	Ambiguous	Critical	Critical

Interpretation Notes:

- **Excellent:** High efficiency, within or at budget. Strong performance.
- **On Track:** Project progressing with minimal value dilution relative to budget consumed.

- **Warning:** Inefficiency (dilution) detected, even if currently within budget. Potential future problems if trend continues.
- **Ambiguous:** Over budget, but spend appears efficient relative to budget consumed. Needs investigation: Is the high efficiency justifying the cost (e.g., scope change, premium value delivered), or masking poor overall cost control?
- **Critical:** Over budget combined with standard or poor efficiency relative to budget consumed. Clear problem with cost control and/or value realization. Requires immediate attention.

Resource Optimization

Our solution addresses the core optimization challenge by recommending consultant assignments that maximize overall value and project health, subject to operational constraints and informed by historical performance data.

Optimization Objective and Constraints

The core goal is to find the assignment matrix A (consultant c, engagement en, hours in week w) that **maximizes predicted value minus switching costs**, subject to operational constraints, including adherence to effective staffing patterns derived from historical data.

1. **Objective Function:** Maximize the anticipated strategic value derived from project performance (predicted SPI, VEC), weighted by strategic importance (w_{en}) and risk (r_{en}), while minimizing the disruption associated with changing consultants' assignments ($pse \times is_switch_{c,en,w}$).

$$\max_{A} \underbrace{\sum_{en \in EN} [w_{en} \times r_{en} \times (\alpha \times SPI_{en} + \beta \times VEC_{en})]}_{\text{Maximize Predicted Value}} - \underbrace{\sum_{c,en,w} pse \times is_switch_{c,en,w}}_{\text{Minimize Switching Cost}}$$

2. **Key Constraint: Adherence to ARR-Adjusted Staffing Ratios (Hard Boundary with Tolerance)** Staffing guidance is implemented as a hard constraint. The model requires the *proposed* staffing ratio for a group (level l, practice pr) on an engagement (en) to stay within a defined **absolute tolerance** (δ) of a **target ratio adjusted by historical Assignment Realization Rate (ARR)**. This anchors the recommendation to efficient historical patterns while allowing defined flexibility.

    - **Defining Terms:** Key inputs like Assignment Realization Rate (ARR) and the detailed methodology for calculating the `TargetRatio_ARR_Adjusted` (using historical actual ratios and ARR) are defined in Appendix 4.
    - **Constraint Logic (Conceptual):** The optimizer must ensure the absolute difference between the proposed staffing ratio (P) and the calculated ARR-adjusted target ratio (T) does not exceed the tolerance δ for each group:

$$|P - T| \leq \delta$$

    (Appendix 4 provides the full mathematical details, including the definition of P and T, and Appendix 2 shows the linearized form used in the MILP implementation).

Managerial Oversight and Strategic Deviation:

This model provides a data-driven recommendation based on optimizing value and stability *within the defined tolerance around ARR-adjusted historical staffing patterns*. It balances technical optimization with practical implementation needs. **Fundamental strategic shifts in staffing mix (e.g., intentionally adding more seniors due to unforeseen complexity, changing the practice mix for a new project phase) fall**

outside this automated guidance and require explicit managerial judgment. The platform provides the optimized baseline reflecting adjusted historical effectiveness; managers use this input, leveraging their expertise to decide when overriding the ratio constraints is necessary for project success.

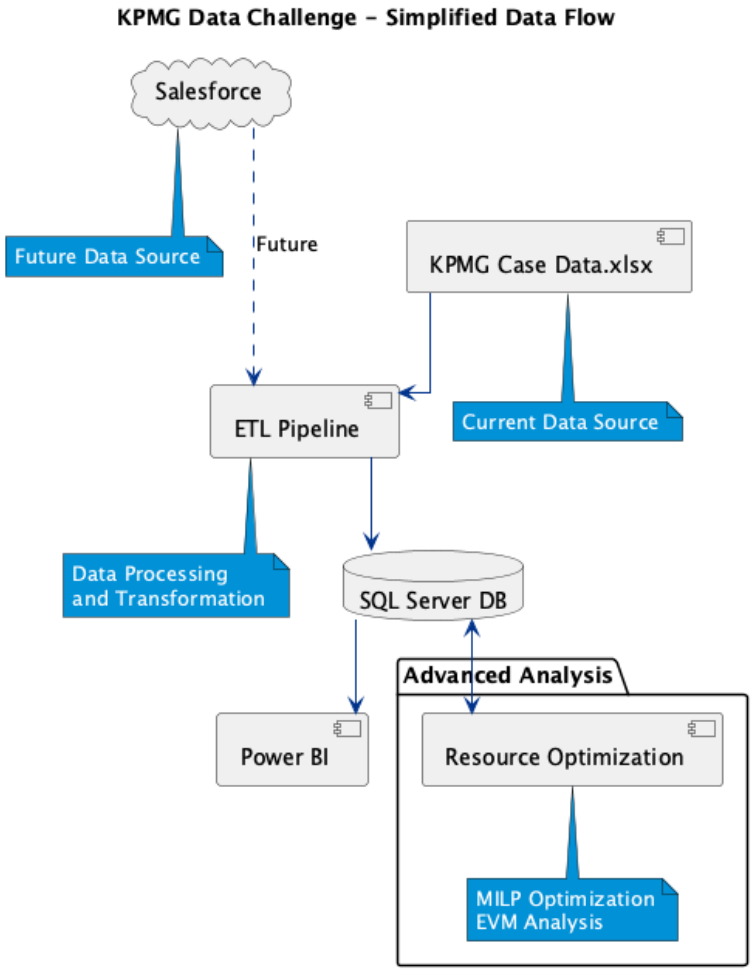
(Note: The MILP implementation detailed in Appendix 2 directly reflects this constrained optimization approach, using the linearized equivalent of the conceptual constraint described above.)

Key Performance Indicators (KPIs)

Based on these business objectives, our solution focuses on the following core KPIs enabling proactive management:

1. **VEC (rolling)**: Value Extraction Coefficient tracking financial realization efficiency.
2. **Budget Burn (rolling)**: Ratio of Actual Cost to Budget ($\frac{\sum AC}{BAC}$) tracking cost control. *Used with VEC for financial health.*
3. **SPI (rolling)**: Schedule Performance Index tracking delivery progress against plan (using adapted EV calculation).
4. **Benching Rate (internal/external)**: Weekly rolling percentage of available capacity *not assigned* to engagements.
5. **Capacity Utilization Rate (internal/external)**: Weekly rolling ratio of *actual billable hours logged* versus standard capacity.

🏗️ Architecture



Technology Stack

- **Data Analysis & Optimization**: Python (Pandas, NumPy, PuLP)
- **Data Storage**: SQL Server database (T-SQL), targeting Azure SQL compatibility.
- **Visualization**: Power BI
- **Integration Reference**: Sample pipeline script (`/scripts/pipeline/pipeline.py`) demonstrates ETL from potential source systems (e.g., Salesforce conceptually) to SQL Server. *Requires adaptation for KPMG's specific environment.*

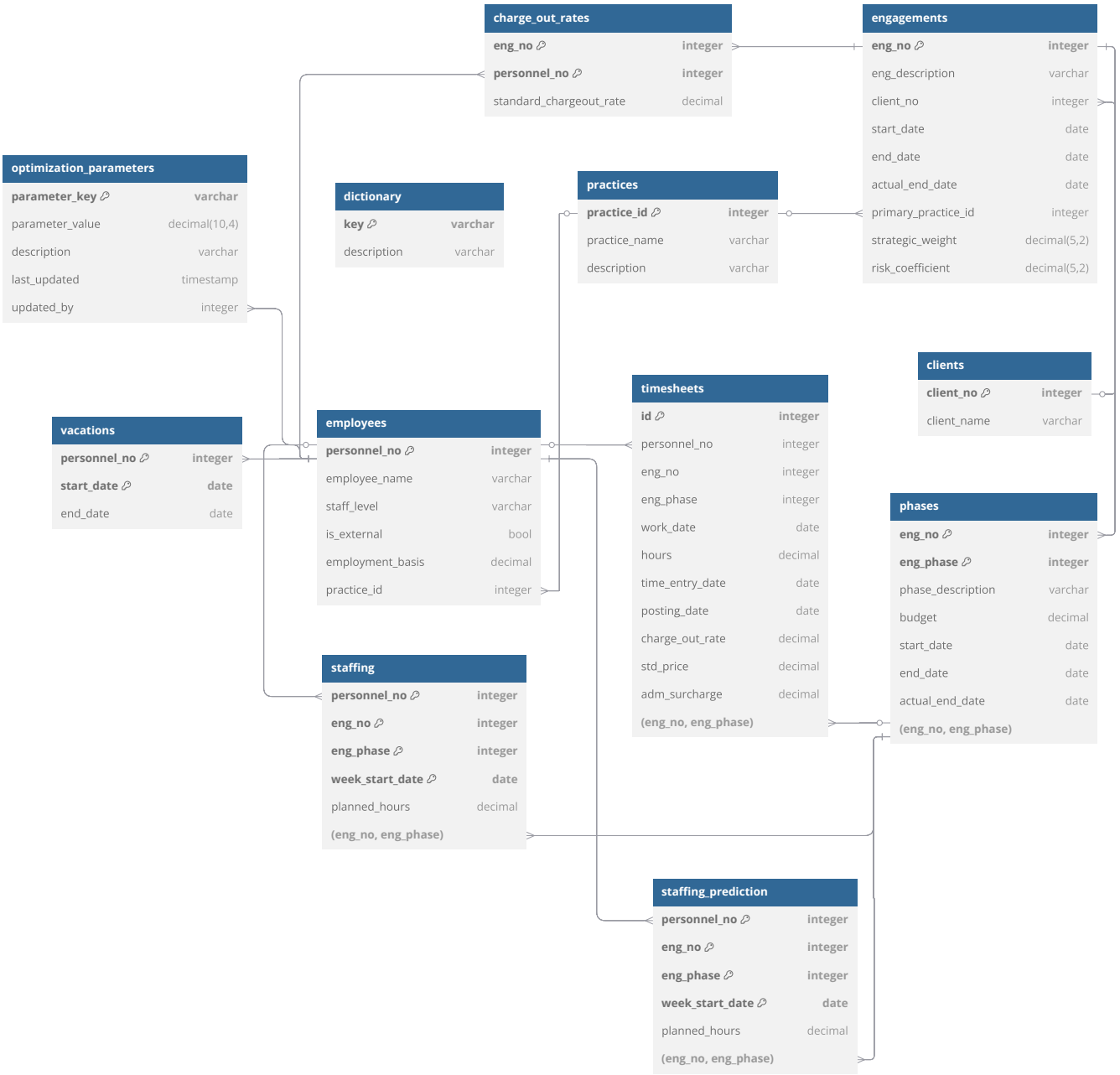
Database Approach

A dedicated database setup is essential for:

1. **Advanced Analytics**: Enabling calculations beyond Power BI's native capabilities.
2. **Complex Optimization**: Providing a structured relational data source for the optimization engine.
3. **Historical Tracking**: Allowing persistent storage for trend analysis and model refinement.
4. **Scalability**: Supporting future expansion of data sources and analytical features.

Data Model

Interactive ERD available here. Static version:



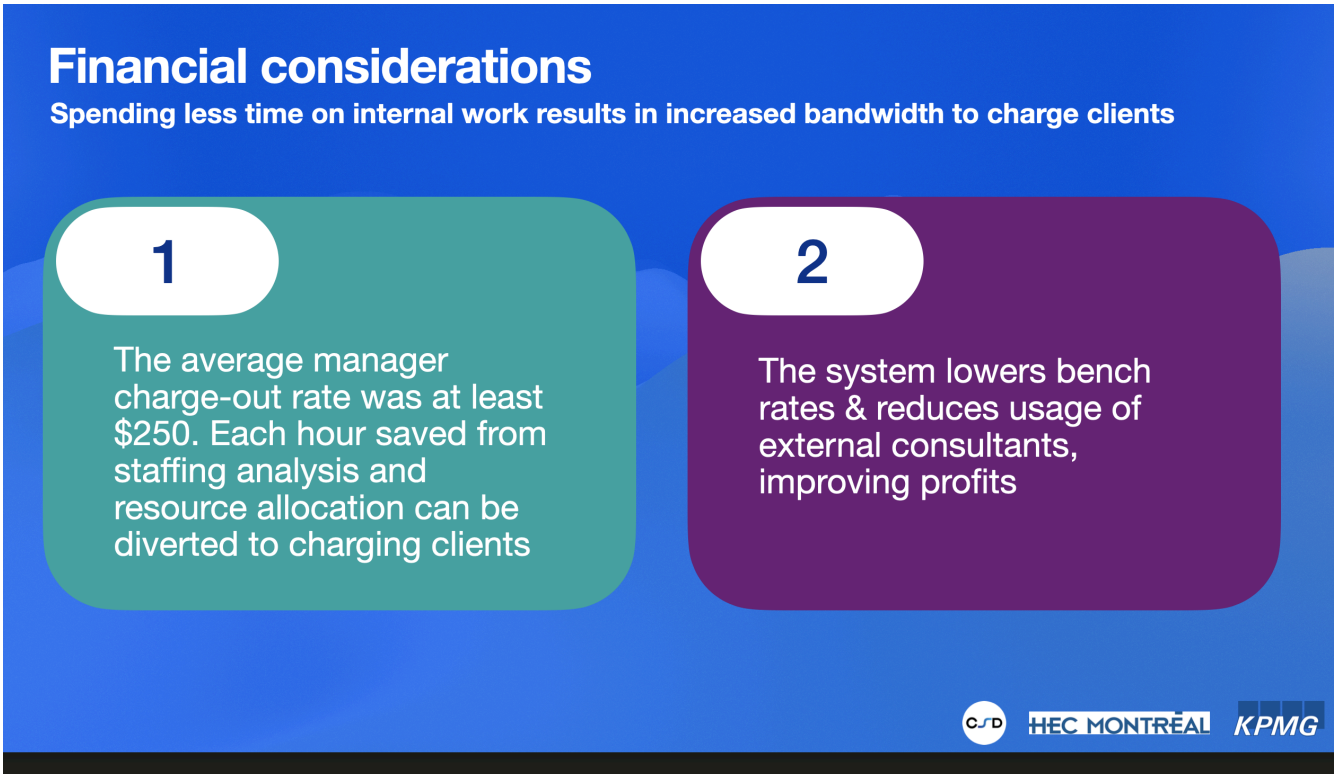
The ERD is implemented in the database, serving as the foundation for the Power BI layer.

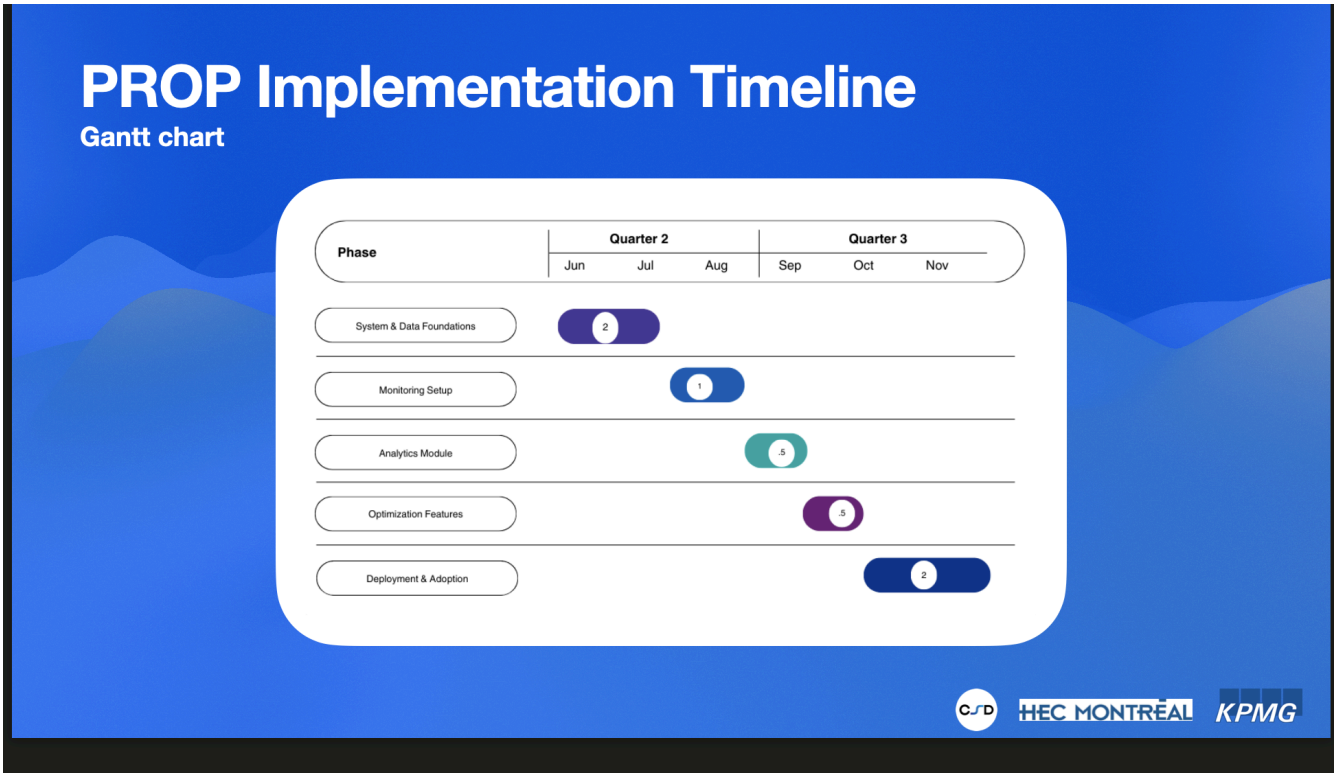
Vacation Tracking Integration

Incorporates a dedicated `vacations` table:

- Stores employee planned time-off.
- Integrates seamlessly with capacity calculations for realistic availability.
- Provides visual conflict indicators in planning views.
- Designed for integration with HR systems (uses synthetic data in PoC).
- Helps avoid resource allocation conflicts due to unaccounted absences.

🏢 Business Recommendations





🛤️ Production Deployment and Dashboard Improvements Suggestions

The current PoC uses synthesized data; production deployment requires integration with actual KPMG data sources.

Synthetic Elements and Production Replacements

PoC Implementation	Recommended Production Data Source	Benefit of Production Data
Vacation Schedules: Generated synthetically	Integration with HR / PTO systems	Accurate capacity planning
Internal/External Status: Derived heuristically	Direct classification from HR / Vendor systems	Precise resource costing and VEC calculation
Employment Basis (Capacity): Assumed standard work week	Actual contracted hours per resource	Accurate capacity for part-time/flexible staff
Engagement/Phase Timelines: Inferred from billing dates	Actual start/end dates from CRM/Project Mgmt	Precise SPI calculation and forecasting
Practice Areas: Defaulted ('SAP')	Actual department/practice assignments (HR/CRM)	Correct staffing mix validation & reporting
Standard Chargeout Rates: Assumed based on level	Official rate cards per level/practice/client	Accurate VEC Rate Efficiency (d_t) calculation
Strategic Weight (w_{en})/Risk (r_{en}): Constant defaults	Defined via business rules/CRM flags/PM input	Optimization reflects true priorities/risk
New Hire Status (eo_c): Not implemented	Flag from HR system based on tenure	More accurate VEC reflecting onboarding period

📝 Implementation Notes

Recommended steps for production deployment:

1. **System Integration**: Establish robust data pipelines (e.g., APIs, scheduled extracts) from source systems (Salesforce, HR, Project Management tools).
2. **Replace Synthetic/Assumed Data**: Populate tables with actual data as outlined above.
3. **Database Deployment**: Implement the SQL schema within KPMG's designated environment (e.g., Azure SQL).
4. **Data Refresh Scheduling**: Configure regular (e.g., daily or weekly) automated data refreshes.
5. **User Access Control**: Implement role-based access security aligned with KPMG policies.
6. **Automated Alert System**: Develop alerts for key KPI thresholds (e.g., low SPI, high bench rate, critical VEC/Burn combination) to enable proactive management.
7. **Data Quality & Volume Assurance:** Ensure sufficient volume and quality of historical data (timesheets, engagements, outcomes, resource details) for effective model calibration and optimization performance validation. **The PoC development highlighted that limitations in the dataset (e.g., a small number of consultants, data entry errors, missing fields) can significantly constrain the practical application and reliability of advanced optimization techniques.** Robust production deployment requires clean, comprehensive data.
8. **Parameter Tuning**: Calibrate optimization parameters ($\alpha, \beta, pse,$\delta$) based on historical data and strategic priorities.

Dashboard Enhancement Roadmap

Recommended enhancements post-deployment:

1. **Historical Performance Integration**: Factor past individual consultant performance metrics (e.g., historical VEC contribution, delivery consistency) into the optimization model.
2. **Skills Taxonomy & Matching**: Implement a detailed skills taxonomy and incorporate skills requirements/matching into the allocation algorithm.
3. **Dynamic Client Priority (w_{en})**: Allow business leaders to dynamically adjust strategic weights for engagements based on evolving priorities.
4. **VEC Deeper Analytics**: Add drill-down views to analyze VEC trends by practice, client sector, engagement type, etc.
5. **Chargeout Discount Monitoring**: Visualize rate realization (d_t) trends and discount levels across engagements and clients.
6. **Refined Switching Penalty (pse)**: Calibrate the switching penalty based on analysis of historical data on productivity impacts of context switching.
7. **Non-Linear Optimization Exploration:** Investigate the use of non-linear solvers or machine learning approaches to handle the conceptual objective function directly, potentially yielding more nuanced results than the linearized demo version.

These steps will evolve the PoC into an indispensable operational tool for resource management and profitability enhancement.

🚀 Getting Started

```
# Clone the repository
git clone [repository-url]

# Navigate to the project directory
cd [repository-directory]

# Install Python dependencies
pip install -r requirements.txt

# Configure the database connection (update connection strings as needed in configuration files
or environment variables)
# Ensure the SQL Server instance is running and the schema (provided) has been deployed

# Open the Power BI dashboard file
open ./dashboards/project_performance.pbix

# Refresh the data connection within Power BI to load data from the database
```

📈 Performance Metrics Example

For a sample engagement phase (Code ending 365, Phase 000010), illustrative metrics observed might include:

- Budget at completion (BAC - Phase): $2.3M
- SPI (Schedule Performance Index): 0.95 *(Example value)*
- VEC (Value Extraction Coefficient): 1.05 *(Example value)*
- Burn (Budget Burn Rate): 0.60 *(Example value)*
- *Other relevant KPIs like Benching Rate and Capacity Utilization for assigned consultants would also be displayed.*

Appendix 1: AI Usage

ChatGPT (free version) and GitHub Copilot were utilized during the development process. Their use included research assistance on technical concepts, brainstorming alternative approaches, generating initial drafts of text content, and providing support for code development and debugging.

Appendix 2: Predicting Optimal Allocation of Employee Resources (MILP Implementation)

This section details the algorithmic approach used for solving the resource optimization problem, specifically outlining the Mixed Integer Linear Programming (MILP) implementation used in the production code. Reference Video Concept.

Implementation Strategy

1. **Data Preparation Phase (Scheduled - Weekly)**
    - **Calculate Engagement Status:** Determine current or projected statuses (SPI, VEC, Budget Burn) for active engagements using methods outlined in Appendix 4.
    - **Retrieve Consultant Availability:** Fetch each consultant's standard capacity (`employees.employment_basis`), adjusting for planned time off (`vacations`) for the target week(s) w.
    - **Calculate Dynamic Target Ratios:** For each active engagement (en), staff level (l), and practice (pr), calculate the *actual* ratio based on `timesheets.hours` from recent weeks, using a weighted 3-week rolling window (weights: 0.6, 0.3, 0.1) to prioritize more recent staffing patterns. *(Note: This serves as the baseline for the simplified constraint in the MILP demo).*
    - **Retrieve Parameters:** Load strategic weights (w_{en}), risk coefficients (r_{en}), and optimization parameters (α, β, pse, δ) from the database or use defaults per Appendix 3.
2. **Optimization Model Formulation (MILP)**
    - **Framework:** Implemented using Python `PuLP`, a library for defining and solving MILP problems.
    - **Decision Variables:**
        - $\text{Hours}_{c,en,ph}$: Planned hours for consultant c on engagement en, phase ph (continuous, non-negative).
    - **Objective Function (Simplified for MILP):**

$$\text{maximize} \sum_{en \in EN} [\alpha \times \text{SPI} + \beta \times \text{VEC}]$$

Where:

- α, β: Weighting coefficients balancing schedule (SPI) vs. value (VEC)
- SPI: Projected SPI contribution calculated as $\text{spi_value} \times \frac{\text{eng_sum_hours}}{40.0}$
- VEC: Projected VEC contribution calculated as $\text{vec_value} \times \frac{\text{eng_sum_hours}}{40.0}$
- pse: None

- **Constraints (Including Simplified Ratio Constraint):**
  - **Consultant Capacity:** Total planned hours for each consultant cannot exceed their adjusted weekly availability:

$$\sum_{en \in EN} \text{Hours}_{c,en,ph} \leq \text{AdjustedCapacity}_c \quad \forall c$$

  - **Staffing Ratio Tolerance (Simplified):** Ensures the planned staffing mix stays within tolerance (δ) of the *previous week's actual ratio* (or the rolling average calculated in prep). This linear constraint approximates the conceptual `StaffingGuidance` term. For each engagement en, level l, practice pr:

    Lower Bound: $\text{PlannedRatio} \geq \text{TargetRatio} \times (1 - \delta)$

    Implemented as:

$$(\text{target_ratio} \times (1 - \delta) - 1) \times \text{level_hours} + \text{target_ratio} \times (1 - \delta) \times \text{other_hours} \leq 0$$

    Upper Bound: $\text{PlannedRatio} \leq \text{TargetRatio} \times (1 + \delta)$

    Implemented as:

$$(1 - \text{target_ratio} \times (1 + \delta)) \times \text{level_hours} - \text{target_ratio} \times (1 + \delta) \times \text{other_hours} \leq 0$$

  - **Switch Logic:** Defines when a consultant starts a new engagement using a "big-M" formulation:

$$\text{hours}_{c,en,ph} \leq M \times \text{is_switch}_{c,en,ph}$$

$$\text{is_switch}_{c,en,ph} \leq \text{hours}_{c,en,ph} \times 100$$

  - **Internal Priority:** Ensures external consultants are only assigned after internal consultants are fully utilized:

$$\text{external_hours} \leq \text{internal_unused} - \text{tolerance}$$

  - **SPI Threshold for Troubled Projects:** For projects with low SPI (< 0.85):

$$\text{project_hours} \geq \text{min_hours_for_troubled}$$

3. **Solver Execution**
   - The PuLP model is passed to a compatible MILP solver (CBC by default) with a time limit of 600 seconds.
   - If an optimal solution is found, the results are processed and returned.
   - If no optimal solution is found, the system performs constraint analysis to diagnose potential infeasibility sources.

4. **Fallback Mechanism & Constraint Analysis**
   - **Constraint Analysis:** If optimization fails, the system analyzes constraints to identify likely sources of infeasibility:
     - Staff level ratio constraints that may be too restrictive
     - Practice ratio constraints with insufficient available consultants
     - Capacity constraints where demand exceeds availability

        - Internal priority constraints creating conflicts
        - Minimum hours requirements for troubled projects that cannot be satisfied
    - **Fallback Strategy:** If optimization is infeasible, the system falls back to the previous week's actual staffing allocation.

5. **Post-Optimization Processing**
    - **Output:** Optimal assignments ($\text{Hours}_{c,en,ph}$) are saved to the `staffing_prediction` table and exported to CSV.
    - **Visualization:** Recommendations are presented on the Power BI dashboard for review by resource managers.
    - **Monitoring & Tuning:** Actual performance is compared to predictions, and optimization parameters ($\alpha, \beta, pse,$\delta$) are adjusted based on results.

Implementation Notes

- **Rolling Ratio Calculation:** The implementation uses a weighted 3-week rolling window for staffing ratios to ensure stability while adapting to recent trends. This ratio serves as the `target_ratio` in the simplified MILP constraints.
- **Vacation Handling:** Available capacity is dynamically adjusted based on planned vacations overlapping with the target week.
- **Normalization Factors:** Projected SPI and VEC contributions are normalized by a 40-hour baseline to maintain consistent scaling.
- **Infeasibility Handling:** The implementation includes robust fallback mechanisms when optimal solutions can't be found, ensuring business continuity.
- **Parameter Tuning:** The tolerance parameter δ is particularly important - higher values provide more flexibility at the cost of potentially deviating further from established staffing patterns. This parameter acts as the simplified proxy for the conceptual `gamma^{\text{dynamic}}`.

Appendix 3: Modelling Assumptions

This appendix outlines the key assumptions underpinning the analysis presented in this document. These assumptions were necessary due to data limitations or for simplification in the Proof of Concept modelling process. They are categorized into critical and supporting assumptions.

Critical Assumptions

1. **Cost Performance Index (CPI) Proxy for EV Calculation is set at 0.98.**
    - **Basis:** In lieu of reliable % Complete data for traditional EV calculation, this PoC uses an efficiency factor (conceptually similar to CPI) applied to Actual Cost to estimate Earned Value (`EV = AC * 0.98`). The 0.98 value is an initial estimate, loosely based on standard project contingency ideas and assumed operational efficiency.
    - **Impact:** Enables calculation of SPI (`SPI = EV/PV`) within the EVM framework adapted for consulting. **Recommendation:** This factor should be refined or replaced with empirically derived efficiency measures based on actual project performance data and potentially linked to VEC trends in a production environment.

Supporting Assumptions

1. **Phase Duration Estimation is Based on Staff Allocation.**
    - **Impact:** Initial project phase duration estimates can be derived by dividing the total budgeted effort (estimated from budget and average rates) by the planned personnel allocation (number of FTEs). This aids in projecting completion dates based on planned resource assignments for PV calculations.
2. **Staffing Level Distribution is Relatively Consistent Across Phases within an Engagement.**
    - **Basis:** Analysis of sample engagement data suggested relatively stable ratios in staff levels (e.g., Consultant, Manager) across different phases of the *same* engagement.
    - **Impact:** Allows for the calculation of reasonably representative weighted average chargeout rates for initial engagement estimations or where phase-specific rates are unavailable.
3. **Project Phases Progress Sequentially for PV Calculation.**
    - **Impact:** Projects are assumed to progress linearly through defined phases without significant overlap for the purpose of calculating Planned Value (PV) based on phase timelines.
4. **Phase Start Date is the First Logged Work Date for that Phase.**
    - **Impact:** In the absence of explicitly defined phase start dates in the dataset, this definition provides a consistent reference point for timeline analysis (Days Elapsed, Schedule % Elapsed).
5. **Client Identity is Uniquely Determined by Client Number, this extends to other conflicts.**
    - **Basis:** Standard database practice; the unique client number serves as the primary key for client identification.
    - **Impact:** Ensures data integrity and enables consistent client tracking and aggregation across engagements.
6. **Staff at Equivalent Levels/Practices are Functionally Interchangeable (Excluding Switching Costs).**
    - **Impact:** For initial modelling, personnel within the same practice and at the same level are considered functionally equivalent regarding task execution capability. The Phase Switching Efficiency factor (`pse`) was set to **1** in the PoC optimization model, **explicitly assuming no productivity loss or time cost associated with a consultant switching between different engagements or phases.** This simplification was made to reduce model complexity for the PoC. In a production environment, `pse` should be calibrated (likely to a value < 1 or represented as a cost penalty) to reflect the realistic overhead and learning curve adjustments associated with context switching.
7. **All Engagements Have Equal Strategic Weight ($w_{en} = 1$) Initially.**
    - **Basis:** Absence of specific prioritization criteria in the initial dataset.
    - **Impact:** Simplifies initial optimization modelling. **Recommendation:** Incorporate actual strategic weights from business input or CRM data.
8. **All Engagements Carry Uniform Risk ($r_{en} = 1$) Initially.**
    - **Basis:** Project-specific risk data was not available for this analysis.
    - **Impact:** Facilitates simplified preliminary analysis. **Recommendation:** Include risk differentiation based on defined criteria or PM assessments.
9. **Negative Hour Logs Represent Adjustments within the Same Client's Portfolio.**
    - **Basis:** Observed data patterns suggest negative timesheet entries often correlate with other positive entries for the same client, potentially representing budget transfers or corrections.
    - **Impact:** Negative hours are treated as adjustments reducing the overall AC for the client/engagement. Further validation with business process owners is recommended.
10. **Time Reporting Timeliness Varies (Heuristic for Internal/External Proxy).**
    - **Basis:** Analysis revealed variations in timesheet submission lag. A heuristic was used in the PoC: consultants with average lag < 3 days were *provisionally* classified as 'external' (excluding senior

managers) for differentiated analysis.
    - **Impact & Recommendation:** This allows exploring potential differences but requires validation and replacement with actual employment status data from HR systems for accurate `ea_c` calculation.
11. **External Consultants Have a 10% Profitability Adjustment Factor ($ea_c = 0.9$).**
    - **Basis:** A preliminary estimate applied due to lack of specific cost differential data.
    - **Impact:** Adjusts the VEC calculation (`ea_c`) to account for potential margin differences. **Recommendation:** Validate and refine this factor based on actual internal cost vs. external vendor rates.
12. **Chargeout Rates are Negotiated Per Engagement.**
    - **Basis:** Data analysis indicates that a consultant's chargeout rate (`timesheets.charge_out_rate`) can be consistent within one engagement but may vary across different engagements they work on.
    - **Impact:** Supports the calculation of the Rate Efficiency Factor (d_t) in VEC using the engagement-specific actual rate compared to the consultant's standard rate.

Appendix 4: Key Metrics Derivation

Preamble on Aggregation: The metrics defined below are often derived from granular data sources like `timesheets`, `phases`, and `employees`. While specific formulas might be shown at a certain level (e.g., per phase), the underlying mathematical logic allows for these metrics to be calculated and analyzed across various hierarchical dimensions by appropriately filtering and aggregating the source data *before* applying the calculation. Common aggregation levels include, but are not limited to: per Consultant, Staff Level, Practice Area, Phase, Engagement, or Client. The specific level of aggregation chosen depends on the analytical question being addressed. The formulas presented illustrate the core calculation, assuming the relevant data has been pre-aggregated to the desired level.

Database Field References for Key Metrics

Current Date Reference for Metrics Calculation

- **Definition**: When calculating metrics requiring a reference to "today" or "current date" (e.g., for calculating progress to date), the system uses the **maximum `timesheets.work_date` available in the database**.
- **Rationale**: This reflects standard dashboard practice, grounding calculations in the most recent data recorded in the system.
- **Implementation**: In SQL queries, "CURRENT_DATE" or equivalent should be derived as `SELECT MAX(work_date) FROM timesheets`.

Budget at Completion (BAC)

- **Database Source**: `phases.budget` (typically at phase level, can be summed for engagement level).
- **Purpose**: Represents the total approved budget for the scope (phase/engagement), serving as the financial baseline.

Weighted Average Chargeout Rate (Estimation Aid)

- **Formula (Example at Phase Level)**:

$$\text{WeightedRate}_{ph} = \sum_{l=1}^{m} (\text{charge_out_rates.standard_chargeout_rate}_{l} \times \frac{\sum \text{staffing.planned_hours}_{l,ph}}{\sum \text{staffing.planned_hours}_{ph}})$$

- **Purpose**: Proof of Concept estimation of the average standard rate for a scope (e.g., phase) based on the *planned* staffing mix by *level*. Used primarily for initial duration/effort estimates when actual billing data is limited.

Staffing Ratio Per Scope (Actual, Historical)

- **Formula (Example for Level l, Phase ph)**:

$$\text{StaffingRatio}_{l,ph} = \frac{\sum \text{timesheets.hours}_{l,ph}}{\sum \text{timesheets.hours}_{ph}}$$

- **Purpose**: Calculates the actual historical ratio of hours contributed by a subgroup (e.g., staff level) to the total hours logged for a specific scope (e.g., phase). Used for deriving staffing ratio targets/baselines in the optimization.

Hours Required / Estimated Duration (Estimation Aids)

- **Formulae (Example at Phase Level)**:

$$\text{Hours}_{\text{required_est},ph} = \frac{\text{phases.budget}_{ph}}{\text{WeightedRate}_{ph}}$$

$$\text{Duration}_{\text{est},ph} = \frac{\text{Hours}_{\text{required_est},ph}}{\sum(\text{staff_count}_{l} \times \text{daily_hours_per_FTE})}$$

- **Purpose**: PoC estimation of total effort (hours) and timeline (duration) for a scope, based on budget and estimated average rates/staffing. Used primarily for PV calculation inputs.

Days Elapsed

- **Formula (Example at Phase Level)**: $\text{DaysElapsed}_{ph} = \text{CURRENT_DATE} - \text{phases.start_date}_{ph}$ (using `MAX(timesheets.work_date)` as CURRENT_DATE)
- **Purpose**: Measures time progression since the scope started.

Percentage Schedule Elapsed

- **Formula (Example at Phase Level)**: $\text{Schedule\%}_{ph} = \frac{\text{DaysElapsed}_{ph}}{(\text{phases.end_date}_{ph} - \text{phases.start_date}_{ph})}$
- **Purpose**: Standardizes schedule progress measurement, used for calculating Planned Value (PV). Assumes linear progress over the planned duration.

Actual Cost (AC)

- **Formula (Aggregated to Scope, e.g., Phase ph, up to current date)**: $AC_{\text{to date},ph} = \sum_{t \in ph,\ \text{to date}} (\text{timesheets.std_price}_{t} + \text{timesheets.adm_surcharge}_{t})$
- **Purpose**: Calculates the cumulative actual expenditure recorded against the scope to date.

Value Extraction Coefficient (VEC)

- **Formula (Aggregated to Scope, e.g., Phase ph)**:

$$VEC_{ph} = \frac{1}{\text{phases.budget}_{ph}} \times \sum_{t \in ph,\ \text{to date}} [(\text{timesheets.std_price}_t + \text{timesheets.adm_surcharge}_t) \times d_t \times ea_c \times$$

Where:

- $d_t = \frac{\text{timesheets.charge_out_rate}_t}{\text{charge_out_rates.standard_chargeout_rate}_c}$ (Rate Efficiency)
- $ea_c = 1 - 0.1 \times \text{employees.is_external}_c$ (External Adjustment)
- $eo_c = 1 - 0.2 \times \text{isNew}_c$ (Onboarding Adjustment - derived based on employee tenure)

- **Purpose**: Measures financial realization efficiency relative to budget for the defined scope.

Budget Burn

- **Formula (Example at Phase Level)**: $\text{Burn}_{ph} = \frac{AC_{\text{to date},ph}}{\text{phases.budget}_{ph}}$
- **Purpose**: Tracks cost consumption relative to the total budget for the scope.

Planned Value (PV)

- **Formula (Example at Phase Level)**: $PV_{ph} = \text{phases.budget}_{ph} \times \text{Schedule\%}_{ph}$
- **Purpose**: Represents the budgeted cost of work scheduled to have been completed by the current date for the scope.

Earned Value (EV) - Adapted PoC Method

- **Formula (PoC Method, Example at Phase Level)**: $EV_{ph} = AC_{\text{to date},ph} \times 0.98$ (using the assumed CPI proxy from Appendix 3)
- **Purpose**: Represents the estimated value of work completed for the scope, **adapted for the consulting context where traditional % Complete is difficult to measure reliably.**

Schedule Performance Index (SPI)

- **Formula (Example at Phase Level)**: $SPI_{ph} = \frac{EV_{ph}}{PV_{ph}}$ (Handle division by zero if PV is zero)
- **Purpose**: Quantifies schedule efficiency (actual progress value vs planned progress value) for the scope. An SPI < 1 indicates being behind schedule.

Weekly Benching Rate

- **Formula (Overall Internal)**: $\text{Benching Rate}_w = (1 - \frac{\sum_{c \in \text{Internal}, en} \text{staffing.planned_hours}_{c,en,w}}{\sum_{c \in \text{Internal}} \text{employees.employment_basis}_c}) \times 100\%$
- **Definition**: Percentage of total available internal consultant capacity ***not assigned*** to specific engagements in week w.
- **Purpose**: KPI measuring unallocated internal resource time.

Weekly Capacity Utilization Rate

- **Formula (Per Consultant c)**: $\text{Capacity Utilization}_{c,w} = \frac{\sum_{en} \text{timesheets.hours}_{c,en,w}}{\text{employees.employment_basis}_c} \times 100\%$ (Can be averaged across groups)
- **Definition**: Percentage of consultant's contracted weekly capacity utilized in ***actual logged billable hours*** during a specific week.
- **Purpose**: KPI measuring productive engagement time relative to capacity.

Weekly Assignment Realization Rate (ARR)

- **Formula (Conceptual for a defined 'group' in week w)**:

$$\text{ARR}_{g,w} = \frac{\sum_g \text{timesheets.hours}_w}{\sum_g \text{staffing.planned_hours}_w}$$

  (Handle division by zero if planned_hours is zero)

- **Definition**: Ratio of *actual hours logged* compared to *planned hours assigned* for a specific group (defined by aggregation dimensions like consultant, level, practice, engagement) over a defined period (typically week w).

- **Purpose**: Internal metric assessing the accuracy of planning/forecasting and providing diagnostic insights for the optimization process (see Resource Optimization section).

ARR-Adjusted Target Staffing Ratio Calculation

- **Definition**: The ARR-adjusted target staffing ratio provides a data-informed baseline for optimized resource allocation, reflecting both historical staffing patterns and their effectiveness.

- **Calculation Steps for Target Ratio (**$\text{TargetRatio}_{l,pr,en,w}^{\text{ARR-Adjusted}}$**)**:

  1. **Historical Averages:** Calculate time-weighted averages of the historical *Actual Ratio* ($\text{AverageActualRatio}_g$) and historical *ARR* ($\text{AverageARR}_g$) for the group g (level l, practice pr) on the engagement over recent relevant periods.
     - $\text{AverageActualRatio}_g = \text{WeightedAverage}(\frac{\text{timesheets.hours}_g}{\text{timesheets.hours}_{en}})$
     - $\text{AverageARR}_g = \text{WeightedAverage}(\frac{\text{timesheets.hours}_g}{\text{staffing.planned_hours}_g})$
  2. **Preliminary Adjustment (Product):** Calculate the preliminary target by multiplying the average actual ratio by the average ARR for the group. This step uses ARR to scale the historical actual ratio based on past planning accuracy/intensity.
     - $\text{PreliminaryTarget}_g = \text{AverageActualRatio}_g \times \text{AverageARR}_g$
  3. **Normalization:** Normalize the preliminary targets across all groups (g') on the engagement to ensure the final target ratios sum to 1.
     - $\text{TargetRatio}_g^{\text{ARR-Adjusted}} = \frac{\text{PreliminaryTarget}_g}{\sum_{\text{all groups } g' \text{ on engagement}} \text{PreliminaryTarget}_{g'}}$

- **Optimization Constraint Formulation:** The optimizer must satisfy the following bounds for each group (level l, practice pr, engagement en, week w):

$$(1-\delta) \times \text{TargetRatio}_{l,pr,en,w}^{\text{ARR-Adjusted}} \leq \text{ProposedRatio}_{l,pr,en,w} \leq (1+\delta) \times \text{TargetRatio}_{l,pr,en,w}^{\text{ARR-Adjusted}}$$

  where $\text{ProposedRatio}_{l,pr,en,w} = \frac{\sum_{c \in C_{l,pr}} \text{Hours}_{c,en,w}}{\sum_{c \in C} \text{Hours}_{c,en,w}}$ and δ is the configurable tolerance parameter (e.g., 0.1 for 10%).

- **Purpose**: This constraint ensures that the optimization model's staffing recommendations adhere to historically effective staffing patterns while allowing for necessary flexibility.

Appendix 5: Mathematical Notation Conventions

This appendix documents the standardized notation used in formulas throughout this document to ensure consistency and clarity.

Suffix Conventions

All variables in formulas use consistent suffixes to denote the entity they refer to:

Suffix	Entity	Example	Description
`c`	Consultant/Employee	ea_c (External Adjustment)	Refers to a specific consultant/employee
`t`	Transaction/Timesheet	AC_t (Actual Cost of transaction)	Refers to a specific timesheet entry
`ph`	Phase	VEC_{ph} (VEC of a phase)	Refers to a specific engagement phase
`en`	Engagement	SPI_{en} (SPI of engagement)	Refers to a specific engagement
`l`	Staff Level	$\text{StaffingRatio}_{l,ph}$	Refers to a specific staff level (e.g., Manager)
`pr`	Practice	$C_{l,pr}$ (Consultants in practice)	Refers to a practice area (e.g., SAP)
`w`	Week	$\text{Hours}_{c,en,w}$ (Weekly hours)	Refers to a specific week (time period)
`g`	Group	$\text{ARR}_{g,w}$ (ARR for a group)	Refers to a specific grouping (e.g., by level, practice, etc.)

Core Variables and Factors

Key Metrics & Financial Variables

Variable	Definition	Source/Calculation Reference
AC_t	Actual Cost (transaction)	`timesheets` fields
AC_{scope}	Actual Cost (scope to date)	Sum ofAC_t for scope
BAC_{scope}	Budget at Completion (scope)	`phases.budget` (or sum)
PV_{scope}	Planned Value (scope)	Appendix 4 formula
EV_{scope}	Earned Value (scope, PoC)	Appendix 4 formula
SPI_{scope}	Schedule Performance Index	$EV_{\text{scope}}/PV_{\text{scope}}$
VEC_{scope}	Value Extraction Coefficient	Appendix 4 formula
$\text{Burn}_{\text{scope}}$	Budget Burn Rate	$AC_{\text{scope}}/BAC_{\text{scope}}$

Adjustment Factors

Factor	Definition	Formula / Source	Purpose
d_t	Rate Efficiency Factor	$\frac{\text{timesheets.charge_out_rate}_t}{\text{charge_out_rates.standard_rate}_c}$	Measures charge-out rate efficiency vs. standard
ea_c	External Adjustment Factor	$1 - 0.1 \times \text{employees.is_external}_c$ (configurable)	Accounts for external consultants' impact on profitability
eo_c	Onboarding/New Hire Adjustment Factor	$1 - 0.2 \times \text{isNew}_c$ (configurable)	Accounts for newer consultants' learning curve
pse	Phase Switching Efficiency Penalty	Configurable parameter	Cost/penalty factor applied when switching engagements (**Note:** Set to 1, implying no penalty/loss, in PoC simplification)

Optimization Parameters

Parameter	Definition	Range/Default / Source	Notes
α	SPI weight in objective	[0, 1], Default: 0.5	From `optimization_parameters` table
β	VEC weight in objective	[0, 1], Default: 0.5	From `optimization_parameters` table
w_{en}	Strategic weight of engagement	≥ 0, Default: 1.0	From `engagements.strategic_weight` or business input
r_{en}	Risk coefficient of engagement	≥ 0, Default: 1.0	From `engagements.risk_coefficient` or risk assessment
pse	Phase Switching Efficiency Penalty Coefficient	Configurable parameter	Applied in objective function for `is_switch` variables
δ	Staffing Ratio Tolerance	[0, 1], e.g., 0.1 (10%)	Parameter defining the allowable deviation band in the hard constraint (Appendix 2)
$\text{TargetRatio}_{l,pr,en,w}^{\text{ARR-Adjusted}}$	ARR-Adjusted Target Staffing Ratio	Calculated from historical `timesheets` & ARR	Baseline ratio used in the staffing ratio constraint. See Resource Optimization section for calc.

Appendix 6: ETL (Extract, Transform, and Load) Pipeline

This section describes the Proof of Concept (PoC) pipeline designed to automate data extraction, transformation, and loading into the target SQL Server database for the KPMG Data Challenge project. While this PoC utilizes CSV and Excel files as data sources, a production implementation would involve direct connections to source systems like Salesforce, HR platforms, and project management tools via APIs or database links.

Getting Started (PoC Instructions)

1. **Install Dependencies:** Ensure all required Python libraries are installed:

   ```
   pip install -r requirements.txt
   ```

2. **Configure Database Connection:** Set up the necessary database connection string (e.g., in a `.env` file or environment variables) for the target SQL Server instance.
3. **Run Unified Pipeline Script:** Execute the main pipeline script, providing paths to input data and output directories:

   ```
   python pipeline.py
   ```

   - This script typically orchestrates the steps below.
4. **Alternatively, Run Pipeline Components Individually:**
   - **Extract/Convert:** If starting from Excel, convert relevant sheets to CSV.

     ```
     python excel_to_csv.py
     ```

   - **Transform:** Apply data cleaning, enrichment, and calculation logic.

     ```
     python data_transformation.py
     ```

   - **Generate Load Scripts:** Create SQL DML statements for inserting/updating data.

     ```
     python DML_writer.py
     ```

5. **Execute DML:** Run the generated `output_dml.sql` script against the target SQL Server database to populate the tables.